R Course
Assignment 2

Introduction

In this assignment, you will debug some R functions for working with DNA sequences.

On ILIAS, you will find `assignment2_functions.R` and `assignment2_demo.R`. The functions file contains faulty code. The demo file runs small tests and shows the expected results.

The Functions

1. `gc_content(seq)` should return the percentage of bases in a DNA sequence that are G or C (e.g. `gc_content("GCGCATAT")` should return 50).

2. `reverse_complement(seq)` should replace A with T, T with A, C with G and G with C, and then reverse the sequence (e.g. `reverse_complement(ÄTGC")` should return `"GCAT"`).

3. `transcribe(dna_seq)` should replace every T by U (e.g. `transcribe(ÄTGCGT")` should return `ÄUGCGU"`).

4. `translate(rna_seq)` should translate an RNA sequence into amino acids, reading three bases (one codon) at a time. Stop and return the result immediately when you reach the stop codon UAA. Only the following six codons need to be supported:

- AUG $\rightarrow$ M
- GCU $\rightarrow$ A
- GGG $\rightarrow$ G
- UUU $\rightarrow$ F
- CCC $\rightarrow$ P
- UAA $\rightarrow$ Stop

Example: `translate(ÄUGGCUUAAGGG")` should return `MMA"`.

5. `hardy_weinberg(p)` should return the genotype frequencies $AA = p^2$, $Aa = 2p(1-p)$ and $aa = (1-p)^2$. `hardy_weinberg(0.6)` should return 0.36, 0.48 and 0.16.

6. `random_dna(length)` should return a random DNA sequence of the requested length, using only A, T, C and G.

Your Task

1. Download both files from ILIAS and run `assignment2_demo.R`. Compare the actual outputs with the expected outputs.

2. Fix the bugs in `assignment2_functions.R`, one function at a time. Add comments above each corrected section explaining the bug. Keep the overall structure and test after every change.

3. Add one additional test case of your own for each function.